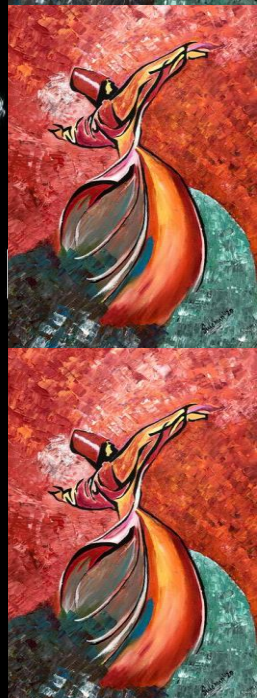




EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



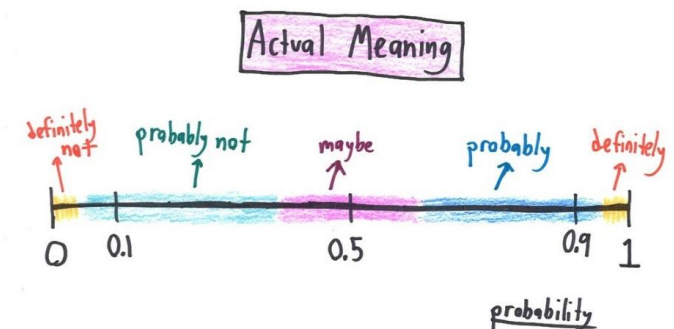


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

Aamir Hasan

Week 7 - Lecture No 12 – 29th September 2025



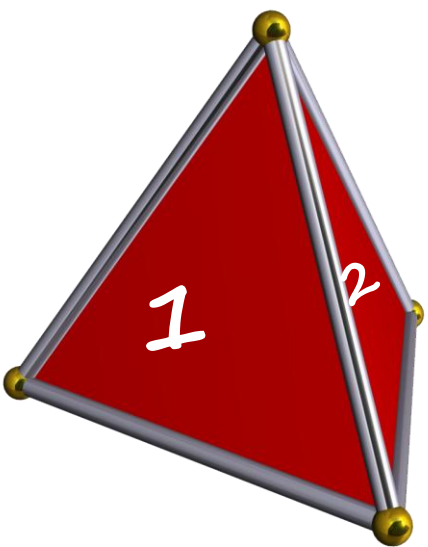
Announcements!

1. Lectures up to No 11 posted
2. Quiz 3 scores posted
3. To enable N-2. we will like to have a quiz No 4 next week?
4. Midterm Exam - October 8, 2025, from 05:40 PM to 06:55 PM – **venue H-507 (L2)** – Good Luck

Agenda for today

1. Unit 4 - Discrete Random Variables

Unit 4
Discrete Random Variables



Equally
likely

A short exercise!

A Random Variable Z_n is defined as $Z_n = \min (X_1; X_2; \dots ; X_n)$, where X_i s are outcomes of identical and independently distributed tetrahedral die (4-sided). As an example $Z_2 = \min (X_1; X_2)$ - meaning Z_2 is the minimum of the outcome of two independent throws of tetrahedral die X_1 and X_2 .

1. What is the $P(Z_{10} > 1)$?

$$P(Z_{10} > 1) = \left(\frac{3}{4}\right)^{10}$$

We define two events

2. What is the $P(Z_{10} = 3)$?

A - Either 3 or 4 appears in ALL ten throws,
B - 4 appears in ALL ten throws

$$P(A) = \left(\frac{2}{4}\right)^{10}$$

$$P(B) = \left(\frac{1}{4}\right)^{10}$$



$$P(Z_{10} = 3) = \left(\frac{2}{4}\right)^{10} - \left(\frac{1}{4}\right)^{10}$$

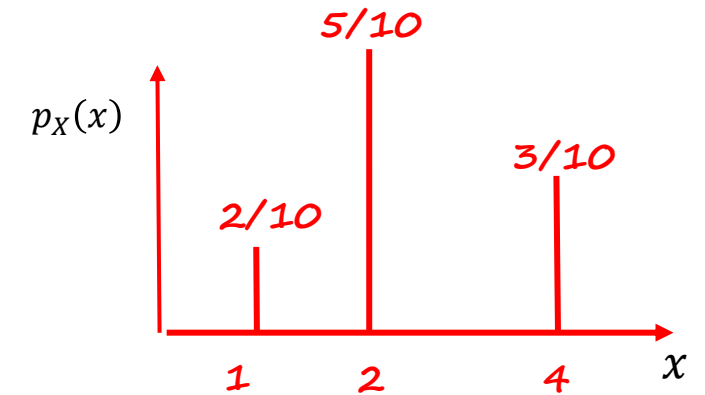
Expectation/mean of a Random Variable

- **Motivation:** Play a game 1000 times.
Random gain at each play described by:

$$X = \begin{cases} 1, & \text{w. p. } 2/10 \sim 200 \\ 2, & \text{w. p. } 5/10 \sim 500 \\ 4, & \text{w. p. } 3/10 \sim 300 \end{cases}$$

- “Average” gain:

$$\rightarrow = \frac{1(200) + 2(500) + 4(300)}{1000} = 1 \cdot \frac{2}{10} + 2 \cdot \frac{5}{10} + 4 \cdot \frac{3}{10}$$



- Definition: $E[X] = \sum_x x p_X(x)$

- **Interpretation:** Average in large number of independent repetitions of the experiment
- **Caution:** If we have an infinite sum it needs to be well-defined.
We assume $\sum_x |x| p_X(x) < \infty$

Expected Value of a Bernoulli Random variable

$$X = \begin{cases} 1, & \text{w.p. } p \\ 0, & \text{w.p. } 1 - p \end{cases} \quad \Rightarrow \quad E[X] = 1(p) + 0(1 - p) = p$$

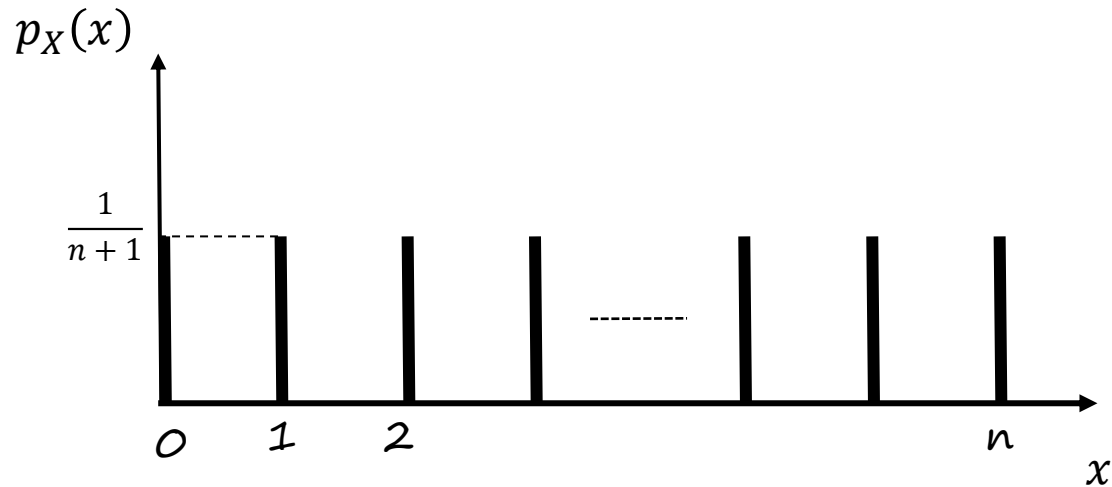
- If X is the Indicator of a event A , $X = I_A$:

$$\Rightarrow X \text{ is } 1 \text{ iff } A \text{ occurs} \quad \Rightarrow p = P(A)$$

$$\Rightarrow E[I_A] = P(A)$$

Expected Value of a Uniform Random variable

- Uniform on 0, 1,, n



- Definition: $E[X] = \sum_x x p_X(x)$

$$\rightarrow E[X] = 0 \frac{1}{n+1} + 1 \frac{1}{n+1} + 2 \frac{1}{n+1} \dots + n \frac{1}{n+1}$$

$$\rightarrow = \frac{1}{n+1} (0+1+ \dots +n) = \frac{1}{n+1} \frac{n(n+1)}{2} = n/2$$

Expected as a population average

- n students
- Weight of i^{th} student: x_i
- Experiment: pick a student at random, all equally likely
- Random variable X : weight of selected student

❖ Assume the x_i are distinct

$$\rightarrow p_X(x_i) = 1/n$$

$$\rightarrow E[X] = \sum_i x_i \frac{1}{n} = \frac{1}{n} \sum_i x_i$$

Properties of Expectations

- Definition: $E[X] = \sum_x x p_X(x)$

- If $X \geq 0$, then $E[X] \geq 0$

→ for all w : $X(w) \geq 0$

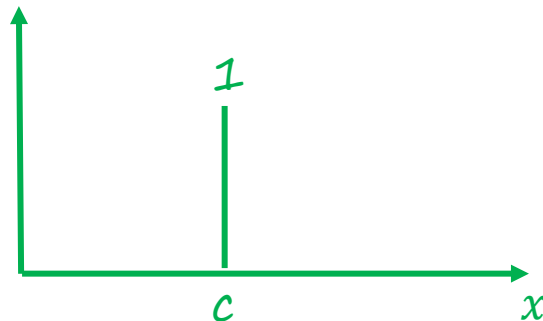
- If $a \leq X \leq b$, then $a \leq E[X] \leq b$

→ for all w : $a \leq X(w) \leq b$

→ $E[X] = \sum_x x p_X(x) \geq \sum_x a p_X(x)$

→ $= a \sum_x p_X(x) = a \cdot 1 = a$

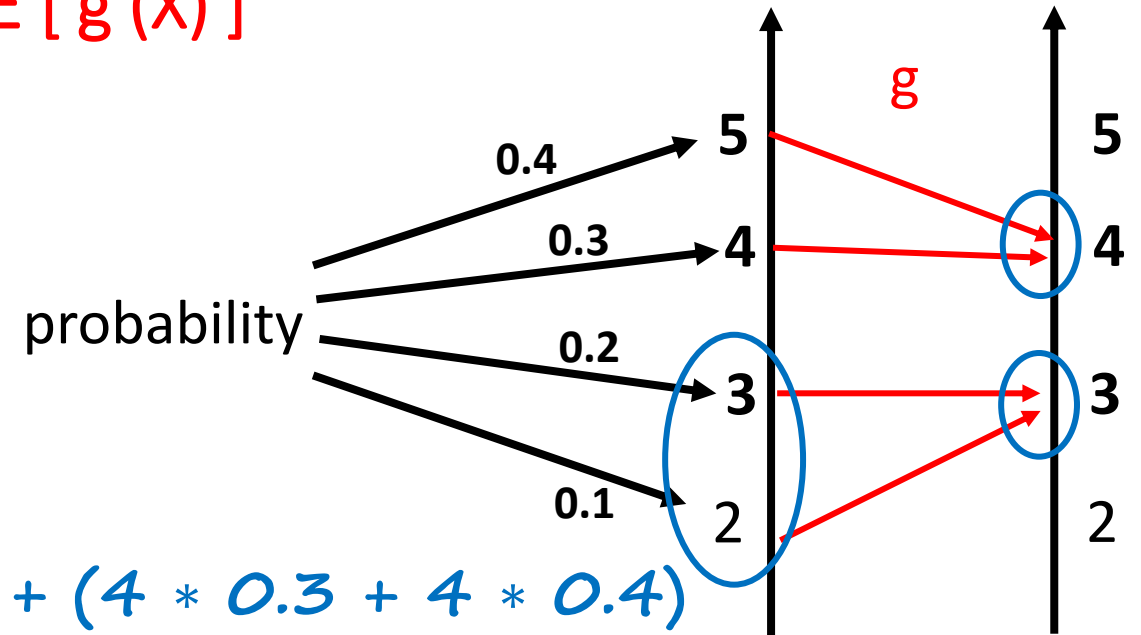
- If c is a constant, $E[c] = c$



→ $E[X] = c \cdot p(c) = c$

The expected value rule, for calculating $E[g(X)]$

- Let X be a r.v. and let $Y = g(X)$
- Averaging over y : $E[Y] = \sum_y y p_Y(y)$
 $3(0.1 + 0.2) + 4(0.3 + 0.4)$



- Averaging over x : $(3 * 0.1 + 3 * 0.2) + (4 * 0.3 + 4 * 0.4)$

$$E[Y] = E[g(X)] = \sum_x g(x) P_X(x)$$

Proof:

$$\sum_y \sum_{x: g(x)=y} g(x) P_X(x)$$

$$= \sum_y \sum_{x: g(x)=y} y P_X(x) = \sum_y y \sum_{x: g(x)=y} P_X(x)$$

$$= \sum_y y P_Y(y) = E[Y]$$

- $E[X^2] = \sum_x x^2 p_X(x)$

$$g(x) = x^2$$

Caution: In general, $E[g(X)] \neq g(E[X])$

$$E[X^2] \neq (E[X])^2$$

Linearity of expectation: $E[aX + b] = aE[X] + b$

→ $X = \text{salary}$ → $E[X] = \text{Average Salary}$

→ $Y = \text{new salary} = 2X + 100$ → $E[Y] = E[2X + 100]$

- Intuitive
- Derivation, based on the expected value rule: → $g(x) = ax + b$
→ $Y = g(X)$

$$\begin{aligned} \rightarrow E[Y] &= \sum_x g(x)p_X(x) \\ &\rightarrow = \sum_x (ax + b)p_X(x) = a \sum_x xp_X(x) + b \underbrace{\sum_x p_X(x)}_1 \\ &\rightarrow = aE[X] + b \end{aligned}$$

$$E[g(X)] = g(E[X])$$

Exceptional g (not always)

Variance – a measure of the spread of a PMF

- Random variable X , with mean $\mu = E[X]$
- Distance from the mean: $X - \mu$?
- Average distance from the mean?

$$E[X - \mu] = E[X] - \mu = \mu - \mu = 0$$

- **Definition of variance:** $\text{var}(X) = E[(X - \mu)^2]$

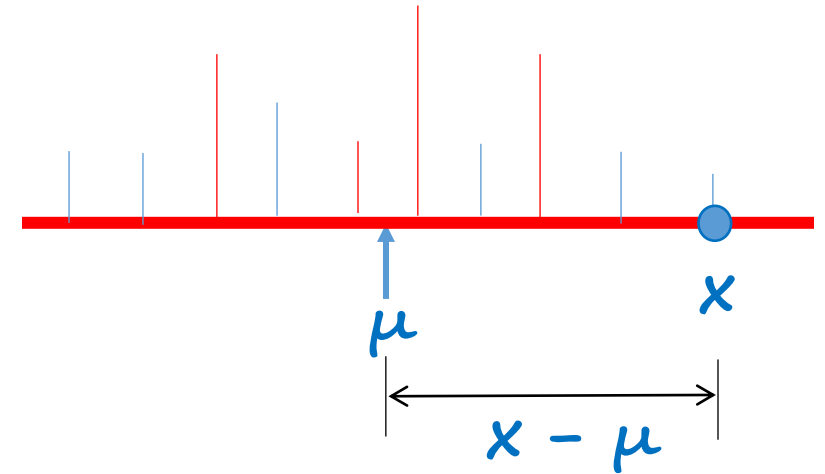
$$\geq 0$$

- Calculation, using the expected value rule, $E[g(X)] = \sum_x g(x)p_X(x)$

$$\text{var}(X) = E[g(X)] = \sum_x (x - \mu)^2 p_X(x)$$

- **Standard deviation:** $\sigma_X = \sqrt{\text{var}(X)}$

2 r.v's Y & X with same mean μ



$$g(x) = (x - \mu)^2$$

Properties of the variance

- Notation: $\mu = E[X]$

$$\text{var}(aX + b) = a^2 \text{var}(X)$$

$$\begin{aligned}\text{var}(3 - 4X)? \\ &= (-4)^2 \text{var}(X) \\ &= 16 \text{var}(X)\end{aligned}$$

- Let $Y = X + b$ $\gamma = E[Y] = \mu + b$

$$\text{var}(Y) = E[(Y - \gamma)^2] = E[(X + \cancel{b} - (\mu + \cancel{b}))^2] = E[(X - \mu)^2] = \text{var}(X)$$

- Let $Y = aX$ $\gamma = E[Y] = a\mu$

$$\text{var}(Y) = E[(aX - a\mu)^2] = E[a^2(X - \mu)^2] = a^2 E[(X - \mu)^2] = a^2 \text{var}(X)$$

$$\text{A useful formula: } \text{var}(X) = E[X^2] - (E[X])^2$$

$$\begin{aligned}\text{var}(X) &= E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2] \\ &= E[X^2] - 2\mu E[X] + \mu^2 = E[X^2] - (E[X])^2\end{aligned}$$

Variance of Bernoulli r.v

A useful formula: $\text{var}(X) = E[X^2] - (E[X])^2$

$$X = \begin{cases} 1, & \text{w.p. } p \\ 0, & \text{w.p. } 1 - p \end{cases}$$

$$E[X] = 1(p) + 0(1 - p) = p$$

$$E[X^2] = 1(p) + 0(1 - p) = p$$

$$\begin{aligned} \text{var}(X) &= E[X^2] - (E[X])^2 \\ &= p - p^2 \\ &= p(1 - p) \end{aligned}$$

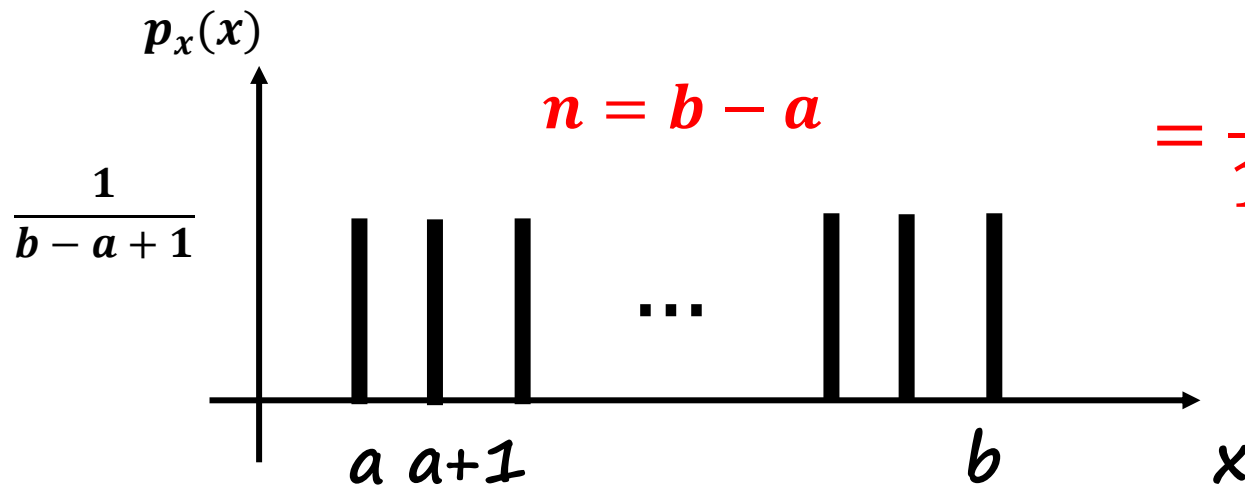
Calculate & plot $\text{Var}(X)$ for different possible values of p

Variance of the uniform



$$\text{var}(X) = E[X^2] - (E[X])^2 = \frac{1}{n+1} \left(0^2 + 1^2 + 2^2 + \dots + n^2 \right) - \left(\frac{n}{2} \right)^2$$

$$\frac{1}{6} n(n+1)(2n+1)$$



$$n = b - a$$

$$= \frac{1}{12} n(n+2)$$

$$\text{Var}(x) = \frac{1}{12} (b-a)(b-a+2)$$